

Лабораторная Работа №14. Статическая маршрутизация в Интернете. Настройка

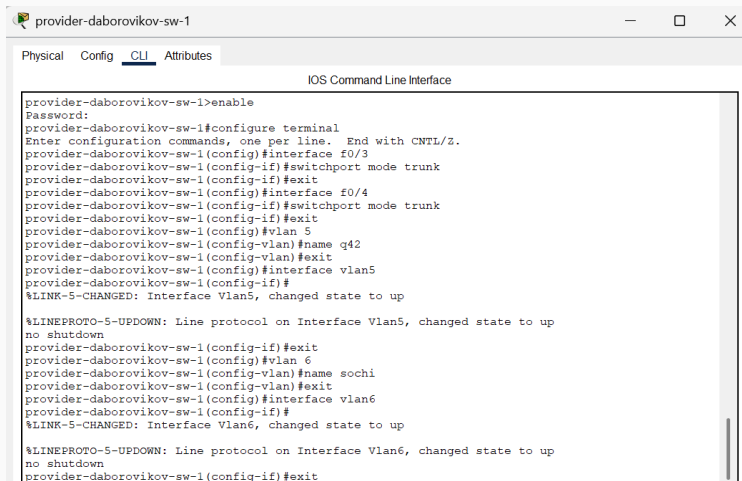
Администрирование локальных сетей

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Настроить взаимодействие через сеть провайдера посредством статической маршрутизации локальной сети организации с сетью основного здания, расположенного в 42-м квартале в Москве, и сетью филиала, расположенного в г. Сочи.



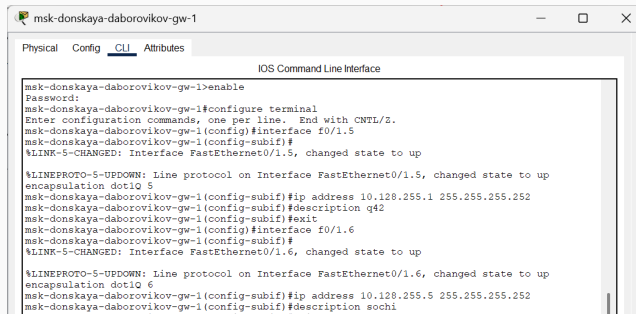
The screenshot shows a window titled 'provider-daborovikov-sw-1' with tabs for 'Physical', 'Config', 'CLI', and 'Attributes'. The 'CLI' tab is active, displaying the 'IOS Command Line Interface'. The interface contains a list of configuration commands for a switch named 'provider-daborovikov-sw-1'. The commands configure the enable password, terminal settings, and two interfaces (f0/3 and f0/4) as trunk ports, each associated with a VLAN (Vlan5 and Vlan6). Vlan5 is named 'q42' and Vlan6 is named 'sochi'. The status of the interfaces is shown as 'up'.

```
provider-daborovikov-sw-1>enable
Password:
provider-daborovikov-sw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
provider-daborovikov-sw-1(config)#interface f0/3
provider-daborovikov-sw-1(config-if)#switchport mode trunk
provider-daborovikov-sw-1(config-if)#exit
provider-daborovikov-sw-1(config)#interface f0/4
provider-daborovikov-sw-1(config-if)#switchport mode trunk
provider-daborovikov-sw-1(config-if)#exit
provider-daborovikov-sw-1(config)#vlan 5
provider-daborovikov-sw-1(config-vlan)#name q42
provider-daborovikov-sw-1(config-vlan)#exit
provider-daborovikov-sw-1(config)#interface vlan5
provider-daborovikov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan5, changed state to up
no shutdown
provider-daborovikov-sw-1(config-if)#exit
provider-daborovikov-sw-1(config)#vlan 6
provider-daborovikov-sw-1(config-vlan)#name sochi
provider-daborovikov-sw-1(config-vlan)#exit
provider-daborovikov-sw-1(config)#interface vlan6
provider-daborovikov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up
no shutdown
provider-daborovikov-sw-1(config-if)#exit
```

Figure 1: Настройка интерфейсов коммутатора provider-daborovikov-sw-1.



```
msk-donskaya-daborovikov-gw-1>enable
Password:
msk-donskaya-daborovikov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#interface f0/1.5
msk-donskaya-daborovikov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up
encapsulation dot1Q 5
msk-donskaya-daborovikov-gw-1(config-subif)#ip address 10.128.255.1 255.255.255.252
msk-donskaya-daborovikov-gw-1(config-subif)#description q42
msk-donskaya-daborovikov-gw-1(config-subif)#exit
msk-donskaya-daborovikov-gw-1(config)#interface f0/1.6
msk-donskaya-daborovikov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.6, changed state to up
encapsulation dot1Q 6
msk-donskaya-daborovikov-gw-1(config-subif)#ip address 10.128.255.5 255.255.255.252
msk-donskaya-daborovikov-gw-1(config-subif)#description sochi
```

Figure 2: Настройка интерфейсов маршрутизатора msk-donskaya-daborovikovgw-1.

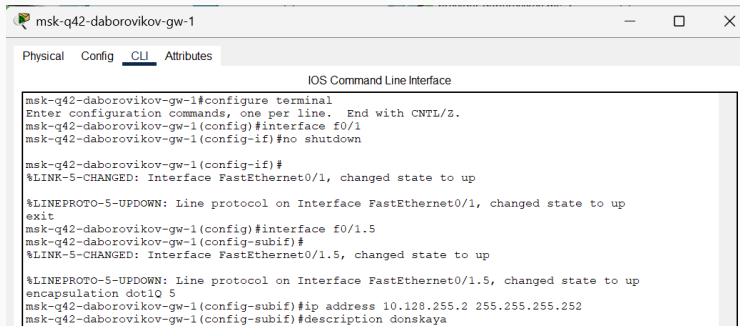
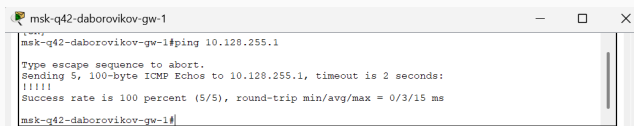


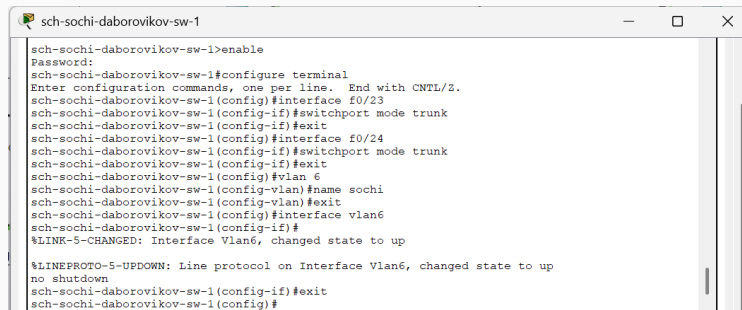
Figure 3: Настройка интерфейсов маршрутизатора msk-q42-daborovikov-gw-1.

Выполнение проверки



```
msk-q42-daborovikov-gw-1
msk-q42-daborovikov-gw-1#ping 10.128.255.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.128.255.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/3/15 ms
msk-q42-daborovikov-gw-1#
```

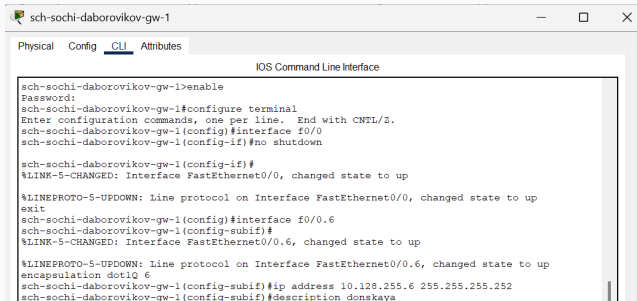
Figure 4: Выполнение проверки



```
sch-sochi-daborovikov-sw-1>enable
Password:
sch-sochi-daborovikov-sw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-daborovikov-sw-1(config)#interface f0/23
sch-sochi-daborovikov-sw-1(config-if)#switchport mode trunk
sch-sochi-daborovikov-sw-1(config-if)#exit
sch-sochi-daborovikov-sw-1(config)#interface f0/24
sch-sochi-daborovikov-sw-1(config-if)#switchport mode trunk
sch-sochi-daborovikov-sw-1(config-if)#exit
sch-sochi-daborovikov-sw-1(config)#vlan 6
sch-sochi-daborovikov-sw-1(config-vlan)#name sochi
sch-sochi-daborovikov-sw-1(config-vlan)#exit
sch-sochi-daborovikov-sw-1(config)#interface vlan6
sch-sochi-daborovikov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up
no shutdown
sch-sochi-daborovikov-sw-1(config-if)#exit
sch-sochi-daborovikov-sw-1(config)#
```

Figure 5: Настройка интерфейсов коммутатора sch-sochi-daborovikov-sw-1.



The screenshot shows a window titled 'sch-sochi-daborovikov-gw-1' with tabs for 'Physical', 'Config', 'CLI', and 'Attributes'. The 'CLI' tab is active, displaying the 'IOS Command Line Interface'. The terminal output shows the following sequence of commands and responses:

```
sch-sochi-daborovikov-gw-1>enable
Password:
sch-sochi-daborovikov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-daborovikov-gw-1(config)#interface f0/0
sch-sochi-daborovikov-gw-1(config-if)#no shutdown

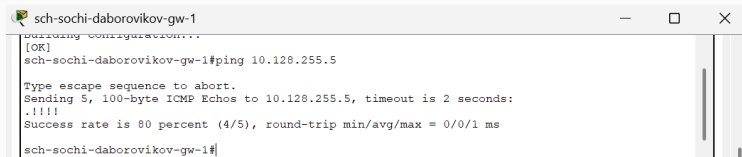
sch-sochi-daborovikov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
sch-sochi-daborovikov-gw-1(config)#interface f0/0.6
sch-sochi-daborovikov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.6, changed state to up
encapsulation dot1Q 6
sch-sochi-daborovikov-gw-1(config-subif)#ip address 10.128.255.6 255.255.255.252
sch-sochi-daborovikov-gw-1(config-subif)#description donskaya
```

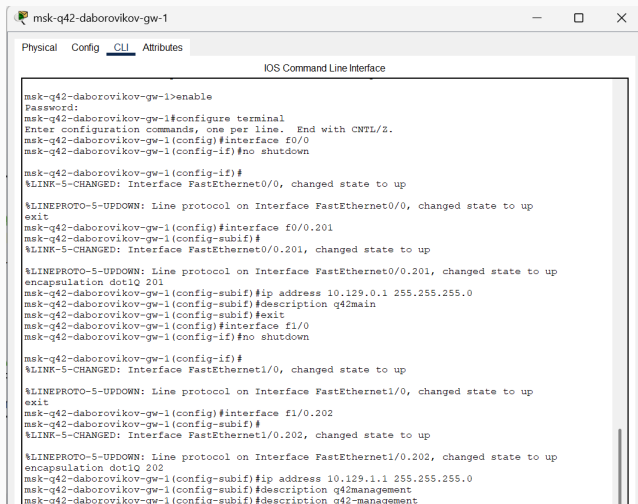
Figure 6: Настройка интерфейсов маршрутизатора sch-sochi-daborovikov-gw-1.

Выполнение проверки.



```
sch-sochi-daborovikov-gw-1
Building configuration...
[OK]
sch-sochi-daborovikov-gw-1#ping 10.128.255.5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.128.255.5, timeout is 2 seconds:
..!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms
sch-sochi-daborovikov-gw-1#
```

Figure 7: Выполнение проверки.



```
msk-q42-daborovikov-gw-1>enable
Password:
msk-q42-daborovikov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-daborovikov-gw-1(config)#interface f0/0
msk-q42-daborovikov-gw-1(config-if)#no shutdown

msk-q42-daborovikov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
msk-q42-daborovikov-gw-1(config)#interface f0/0.201
msk-q42-daborovikov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.201, changed state to up

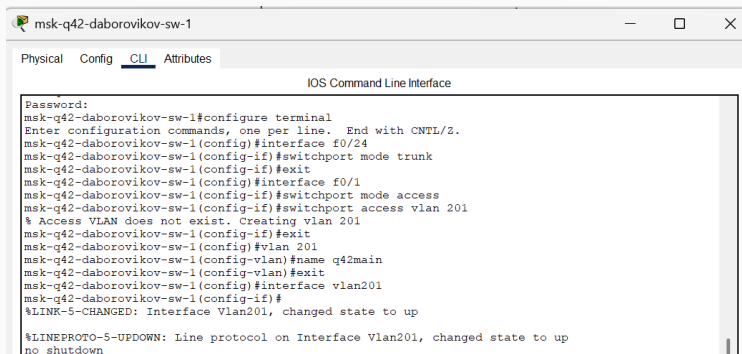
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.201, changed state to up
encapsulation dot1Q 201
msk-q42-daborovikov-gw-1(config-subif)#ip address 10.129.0.1 255.255.255.0
msk-q42-daborovikov-gw-1(config-subif)#description q42main
msk-q42-daborovikov-gw-1(config-subif)#exit
msk-q42-daborovikov-gw-1(config)#interface f1/0
msk-q42-daborovikov-gw-1(config-if)#no shutdown

msk-q42-daborovikov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
exit
msk-q42-daborovikov-gw-1(config)#interface f1/0.202
msk-q42-daborovikov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet1/0.202, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0.202, changed state to up
encapsulation dot1Q 202
msk-q42-daborovikov-gw-1(config-subif)#ip address 10.129.1.1 255.255.255.0
msk-q42-daborovikov-gw-1(config-subif)#description q42management
msk-q42-daborovikov-gw-1(config-subif)#description q42-management
```

Figure 8: Настройка интерфейсов маршрутизатора msk-q42-daborovikov-gw-1.



The screenshot shows a window titled "msk-q42-daborovikov-sw-1" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface". The terminal output shows the following commands and responses:

```
Password:
msk-q42-daborovikov-sw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-daborovikov-sw-1(config)#interface f0/24
msk-q42-daborovikov-sw-1(config-if)#switchport mode trunk
msk-q42-daborovikov-sw-1(config-if)#exit
msk-q42-daborovikov-sw-1(config)#interface f0/1
msk-q42-daborovikov-sw-1(config-if)#switchport mode access
msk-q42-daborovikov-sw-1(config-if)#switchport access vlan 201
% Access VLAN does not exist. Creating vlan 201
msk-q42-daborovikov-sw-1(config-if)#exit
msk-q42-daborovikov-sw-1(config)#vlan 201
msk-q42-daborovikov-sw-1(config-vlan)#name q42main
msk-q42-daborovikov-sw-1(config-vlan)#exit
msk-q42-daborovikov-sw-1(config)#interface vlan201
msk-q42-daborovikov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan201, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan201, changed state to up
no shutdown
```

Figure 9: Настройка интерфейсов коммутатора msk-q42-daborovikov-sw-1.

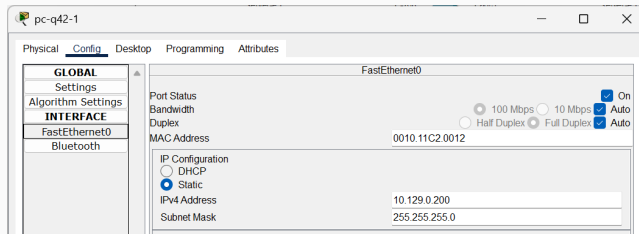
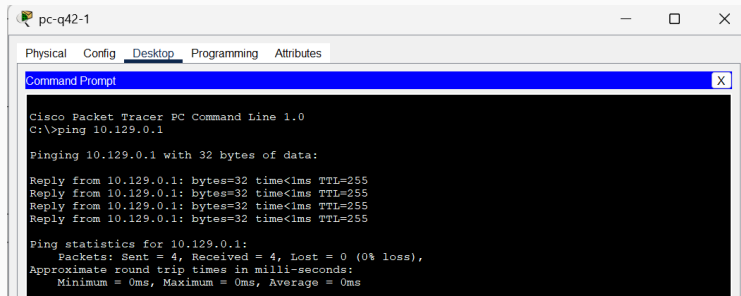


Figure 10: Присвоение адресов оконечному устройству pc-q42-1.



The screenshot shows a window titled "pc-q42-1" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The text in the Command Prompt is as follows:

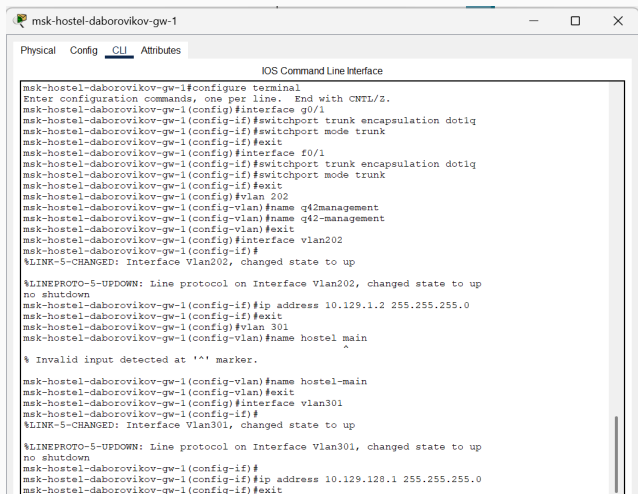
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.129.0.1

Pinging 10.129.0.1 with 32 bytes of data:

Reply from 10.129.0.1: bytes=32 time<1ms TTL=255
Reply from 10.129.0.1: bytes=32 time<1ms TTL=255
Reply from 10.129.0.1: bytes=32 time<1ms TTL=255
Reply from 10.129.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.129.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 11: Выполнение проверки.



The screenshot shows a terminal window titled "msk-hostel-daborovikov-gw-1" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface". The terminal shows the following commands and their outputs:

```
msk-hostel-daborovikov-gw-1#configure terminal
msk-hostel-daborovikov-gw-1(config)#interface g0/1
msk-hostel-daborovikov-gw-1(config-if)#switchport trunk encapsulation dot1q
msk-hostel-daborovikov-gw-1(config-if)#switchport mode trunk
msk-hostel-daborovikov-gw-1(config-if)#exit
msk-hostel-daborovikov-gw-1(config)#interface f0/1
msk-hostel-daborovikov-gw-1(config-if)#switchport trunk encapsulation dot1q
msk-hostel-daborovikov-gw-1(config-if)#switchport mode trunk
msk-hostel-daborovikov-gw-1(config-if)#exit
msk-hostel-daborovikov-gw-1(config)#vlan 202
msk-hostel-daborovikov-gw-1(config-vlan)#name q42management
msk-hostel-daborovikov-gw-1(config-vlan)#name q42-management
msk-hostel-daborovikov-gw-1(config-vlan)#exit
msk-hostel-daborovikov-gw-1(config)#interface vlan202
msk-hostel-daborovikov-gw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan202, changed state to up

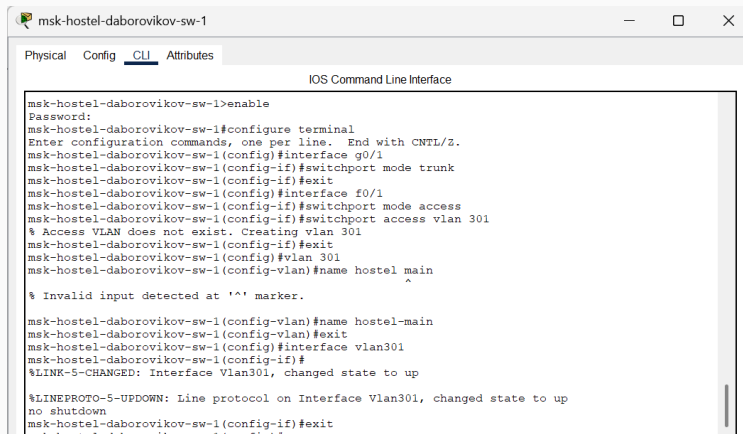
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan202, changed state to up
no shutdown
msk-hostel-daborovikov-gw-1(config-if)#ip address 10.129.1.2 255.255.255.0
msk-hostel-daborovikov-gw-1(config-if)#exit
msk-hostel-daborovikov-gw-1(config)#vlan 301
msk-hostel-daborovikov-gw-1(config-vlan)#name hostel main
msk-hostel-daborovikov-gw-1(config-vlan)#exit
% Invalid input detected at '^' marker.

msk-hostel-daborovikov-gw-1(config-vlan)#name hostel-main
msk-hostel-daborovikov-gw-1(config-vlan)#exit
msk-hostel-daborovikov-gw-1(config)#interface vlan301
msk-hostel-daborovikov-gw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan301, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up
no shutdown
msk-hostel-daborovikov-gw-1(config-if)#
msk-hostel-daborovikov-gw-1(config-if)#ip address 10.129.128.1 255.255.255.0
msk-hostel-daborovikov-gw-1(config-if)#exit
```

Figure 12: Настройка интерфейсов маршрутизирующего коммутатора mskhostel-daborovikov-gw-1.

Выполнение проверки.



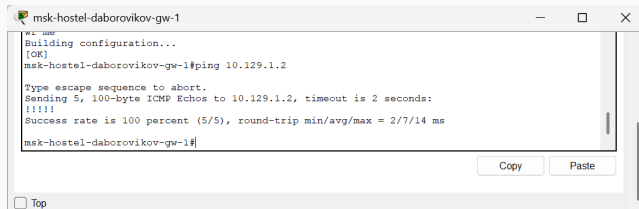
```
msk-hostel-daborovikov-sw-1
Physical Config CLI Attributes
IOS Command Line Interface

msk-hostel-daborovikov-sw-1>enable
Password:
msk-hostel-daborovikov-sw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-hostel-daborovikov-sw-1(config)#interface g0/1
msk-hostel-daborovikov-sw-1(config-if)#switchport mode trunk
msk-hostel-daborovikov-sw-1(config-if)#exit
msk-hostel-daborovikov-sw-1(config)#interface f0/1
msk-hostel-daborovikov-sw-1(config-if)#switchport mode access
msk-hostel-daborovikov-sw-1(config-if)#switchport access vlan 301
% Access VLAN does not exist. Creating vlan 301
msk-hostel-daborovikov-sw-1(config-if)#exit
msk-hostel-daborovikov-sw-1(config)#vlan 301
msk-hostel-daborovikov-sw-1(config-vlan)#name hostel main
^
% Invalid input detected at '^' marker.

msk-hostel-daborovikov-sw-1(config-vlan)#name hostel-main
msk-hostel-daborovikov-sw-1(config-vlan)#exit
msk-hostel-daborovikov-sw-1(config)#interface vlan301
msk-hostel-daborovikov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan301, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up
no shutdown
msk-hostel-daborovikov-sw-1(config-if)#exit
```

Figure 13: Выполнение проверки.



```
msk-hostel-daborovikov-gw-1
Building configuration...
[OK]
msk-hostel-daborovikov-gw-1#ping 10.129.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.129.1.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/7/14 ms
msk-hostel-daborovikov-gw-1#
```

Copy Paste

☐ Top

Figure 14: Настройка интерфейсов коммутатора msk-hostel-sw-1.

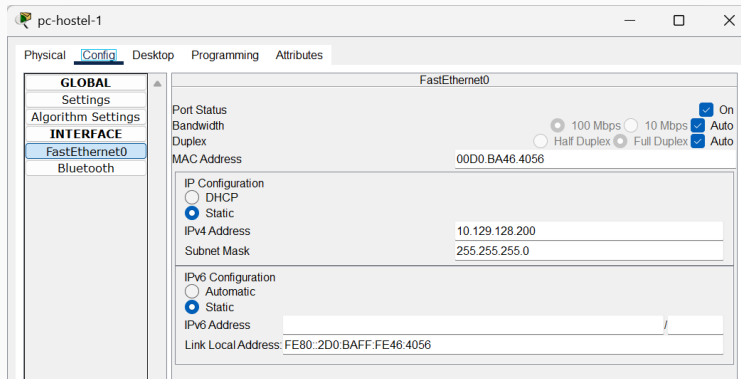
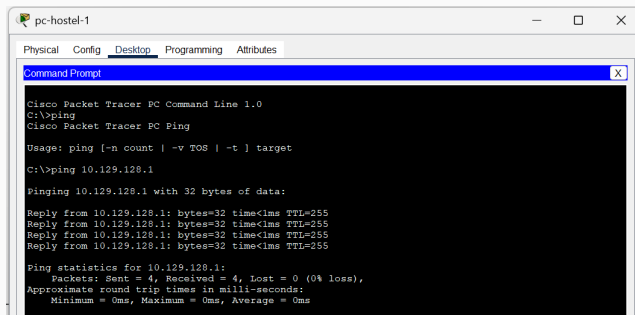


Figure 15: Присвоение адресов оконечному устройству pc-hostel-1.

Выполнение проверки.



The screenshot shows a window titled "pc-hostel-1" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The text in the Command Prompt is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping
Cisco Packet Tracer PC Ping

Usage: ping [-n count | -v TOS | -t ] target

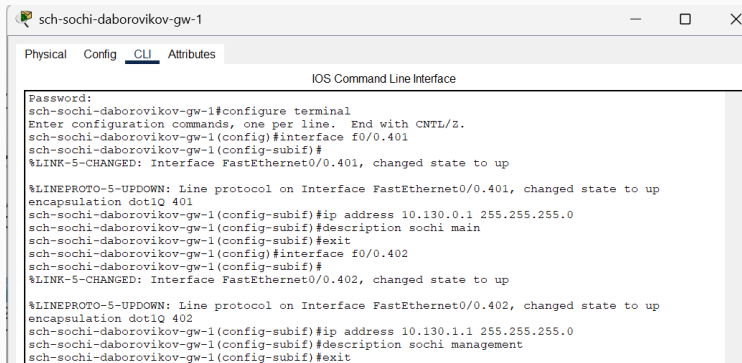
C:\>ping 10.129.128.1

Pinging 10.129.128.1 with 32 bytes of data:

Reply from 10.129.128.1: bytes=32 time<1ms TTL=255
Reply from 10.129.128.1: bytes=32 time<1ms TTL=255
Reply from 10.129.128.1: bytes=32 time<1ms TTL=255
Reply from 10.129.128.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.129.128.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 16: Выполнение проверки.



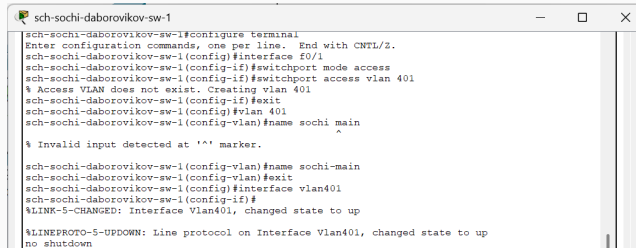
The screenshot shows a web-based configuration interface for a device named 'sch-sochi-daborovikov-gw-1'. The 'CLI' tab is selected, displaying the 'IOS Command Line Interface'. The interface shows a series of configuration commands being entered and executed. The commands configure two interfaces, f0/0.401 and f0/0.402, with IP addresses, descriptions, and encapsulation. The output of the commands is displayed below the input lines.

```
sch-sochi-daborovikov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-daborovikov-gw-1(config)#interface f0/0.401
sch-sochi-daborovikov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.401, changed state to up
encapsulation dot1Q 401
sch-sochi-daborovikov-gw-1(config-subif)#ip address 10.130.0.1 255.255.255.0
sch-sochi-daborovikov-gw-1(config-subif)#description sochi main
sch-sochi-daborovikov-gw-1(config-subif)#exit
sch-sochi-daborovikov-gw-1(config)#interface f0/0.402
sch-sochi-daborovikov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.402, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.402, changed state to up
encapsulation dot1Q 402
sch-sochi-daborovikov-gw-1(config-subif)#ip address 10.130.1.1 255.255.255.0
sch-sochi-daborovikov-gw-1(config-subif)#description sochi management
sch-sochi-daborovikov-gw-1(config-subif)#exit
```

Figure 17: Первоначальная настройка маршрутизатора sch-sochi-daborovikov-gw-1.



```
sch-sochi-daborovikov-sw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-daborovikov-sw-1(config)#interface f0/1
sch-sochi-daborovikov-sw-1(config-if)#switchport mode access
sch-sochi-daborovikov-sw-1(config-if)#switchport access vlan 401
% Access VLAN does not exist. Creating vlan 401
sch-sochi-daborovikov-sw-1(config-if)#exit
sch-sochi-daborovikov-sw-1(config)#vlan 401
sch-sochi-daborovikov-sw-1(config-vlan)#name sochi main
^
% Invalid input detected at '^' marker.

sch-sochi-daborovikov-sw-1(config-vlan)#name sochi-main
sch-sochi-daborovikov-sw-1(config-vlan)#exit
sch-sochi-daborovikov-sw-1(config)#interface vlan401
sch-sochi-daborovikov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan401, changed state to up
no shutdown
```

Figure 18: Первоначальная настройка коммутатора sch-sochi-daborovikov-sw-1.

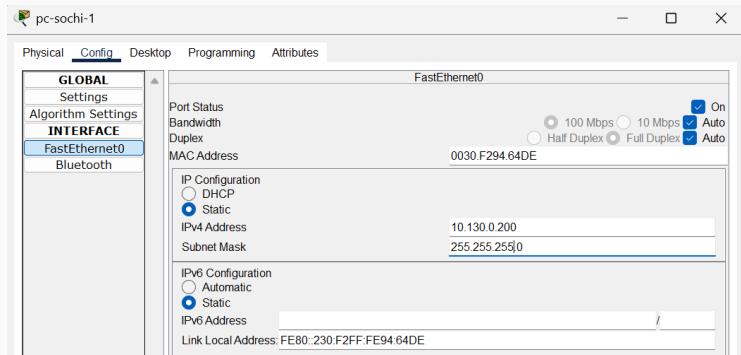
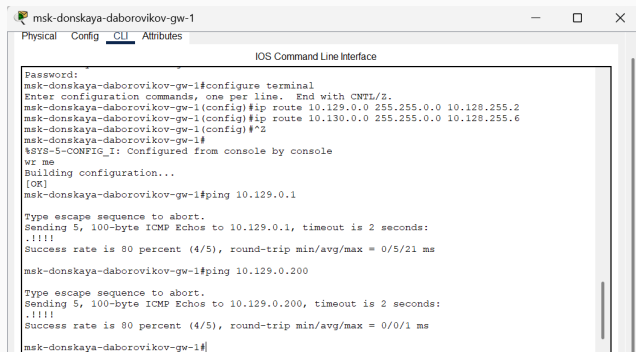


Figure 19: Присвоение адресов оконечному устройству pc-sochi-1.



```
msk-donskaya-daborovikov-gw-1
Physical Config CLI Attributes
IOS Command Line Interface
Password:
msk-donskaya-daborovikov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#ip route 10.129.0.0 255.255.0.0 10.128.255.2
msk-donskaya-daborovikov-gw-1(config)#ip route 10.130.0.0 255.255.0.0 10.128.255.6
msk-donskaya-daborovikov-gw-1(config)#^Z
msk-donskaya-daborovikov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr me
Building configuration...
[OK]
msk-donskaya-daborovikov-gw-1#ping 10.129.0.1

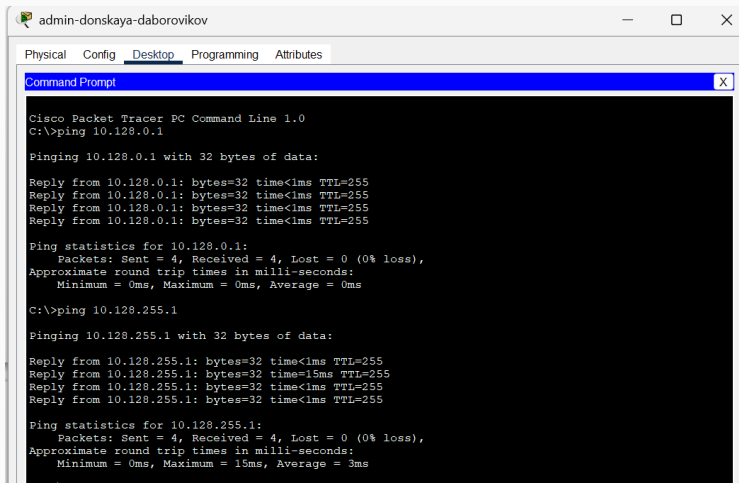
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.129.0.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/5/21 ms

msk-donskaya-daborovikov-gw-1#ping 10.129.0.200

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.129.0.200, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

msk-donskaya-daborovikov-gw-1#
```

Figure 20: Настройка маршрутизатора msk-donskaya-daborovikov-gw-1.



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'admin-donskaya-daborovikov'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' selected. The Command Prompt shows the following commands and output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.0.1

Pinging 10.128.0.1 with 32 bytes of data:

Reply from 10.128.0.1: bytes=32 time<1ms TTL=255
Reply from 10.128.0.1: bytes=32 time<1ms TTL=255
Reply from 10.128.0.1: bytes=32 time<1ms TTL=255
Reply from 10.128.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.128.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

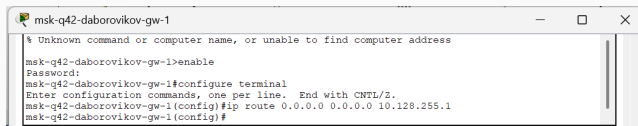
C:\>ping 10.128.255.1

Pinging 10.128.255.1 with 32 bytes of data:

Reply from 10.128.255.1: bytes=32 time<1ms TTL=255
Reply from 10.128.255.1: bytes=32 time=15ms TTL=255
Reply from 10.128.255.1: bytes=32 time<1ms TTL=255
Reply from 10.128.255.1: bytes=32 time<1ms TTL=255

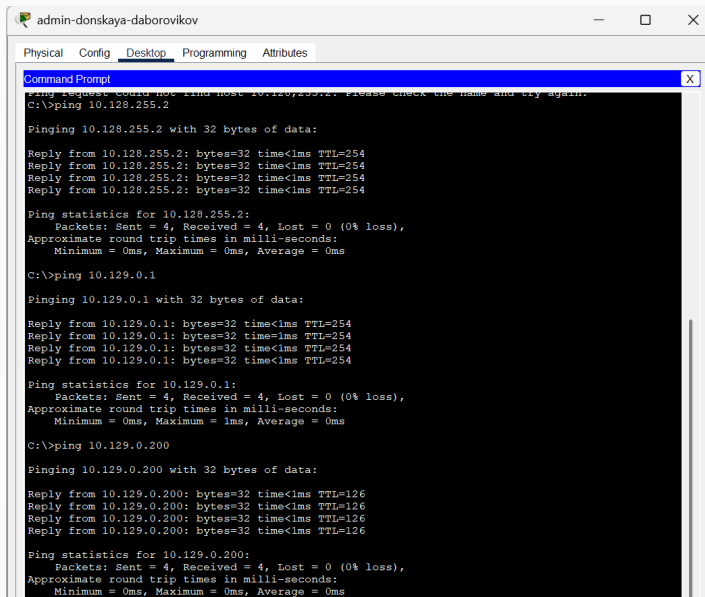
Ping statistics for 10.128.255.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 15ms, Average = 3ms
```

Figure 21: Выполнение проверки.



```
msk-q42-daborovikov-gw-1
% Unknown command or computer name, or unable to find computer address
msk-q42-daborovikov-gw-1>enable
Password:
msk-q42-daborovikov-gw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-daborovikov-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.1
msk-q42-daborovikov-gw-1(config)#
```

Figure 22: Настройка маршрутизатора msk-q42-daborovikov-gw-1.



```
admin-donskaya-daborovikov
Physical Config Desktop Programming Attributes
Command Prompt
ping request could not find host 10.128.255.2. Please check the name and try again.
C:\>ping 10.128.255.2

Pinging 10.128.255.2 with 32 bytes of data:

Reply from 10.128.255.2: bytes=32 time<1ms TTL=254
Reply from 10.128.255.2: bytes=32 time<1ms TTL=254
Reply from 10.128.255.2: bytes=32 time<1ms TTL=254
Reply from 10.128.255.2: bytes=32 time<1ms TTL=254

Ping statistics for 10.128.255.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.129.0.1

Pinging 10.129.0.1 with 32 bytes of data:

Reply from 10.129.0.1: bytes=32 time<1ms TTL=254
Reply from 10.129.0.1: bytes=32 time<1ms TTL=254
Reply from 10.129.0.1: bytes=32 time<1ms TTL=254
Reply from 10.129.0.1: bytes=32 time<1ms TTL=254

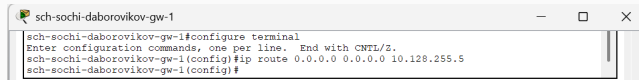
Ping statistics for 10.129.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.129.0.200

Pinging 10.129.0.200 with 32 bytes of data:

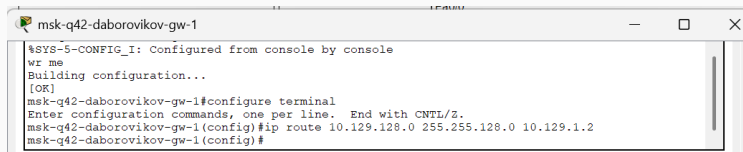
Reply from 10.129.0.200: bytes=32 time<1ms TTL=126
Reply from 10.129.0.200: bytes=32 time<1ms TTL=126
Reply from 10.129.0.200: bytes=32 time<1ms TTL=126
Reply from 10.129.0.200: bytes=32 time<1ms TTL=126

Ping statistics for 10.129.0.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```



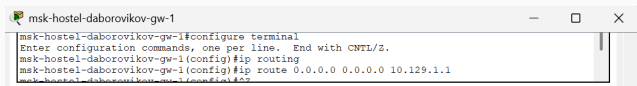
```
sch-sochi-daborovikov-gw-1
sch-sochi-daborovikov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-daborovikov-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.5
sch-sochi-daborovikov-gw-1(config)#
```

Figure 24: Настройка маршрутизатора sch-sochi-daborovikov-gw-1.



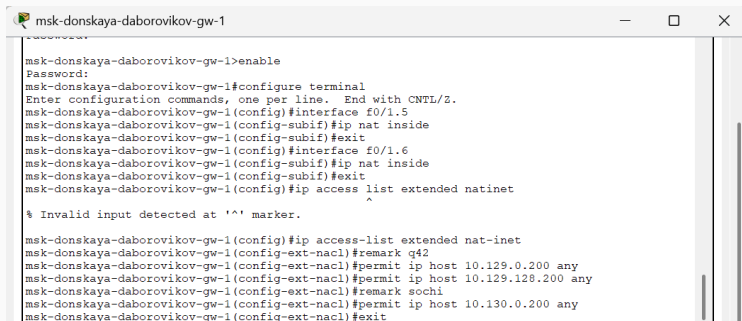
```
msk-q42-daborovikov-gw-1
%SYS-5-CONFIG_I: Configured from console by console
wr me
Building configuration...
[OK]
msk-q42-daborovikov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-daborovikov-gw-1(config)#ip route 10.129.128.0 255.255.128.0 10.129.1.2
msk-q42-daborovikov-gw-1(config)#
```

Figure 25: Настройка маршрутизатора msk-q42-daborovikov-gw-1.



```
msk-hostel-daborovikov-gw-1
msk-hostel-daborovikov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-hostel-daborovikov-gw-1(config)#ip routing
msk-hostel-daborovikov-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.129.1.1
msk-hostel-daborovikov-gw-1(config)#
```

Figure 26: Настройка интерфейсов маршрутизирующего коммутатора mskhostel-daborovikov-gw-1.

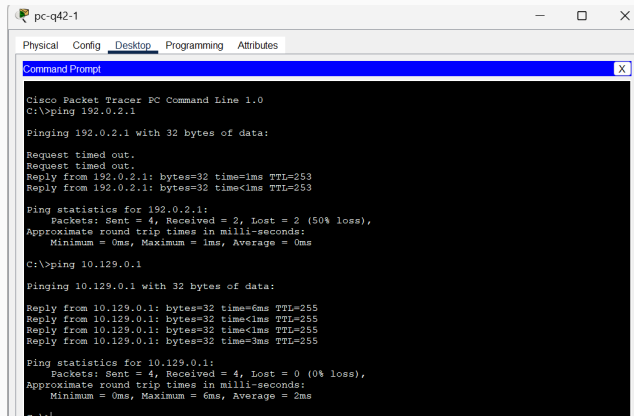


```
msk-donskaya-daborovikov-gw-1>enable
Password:
msk-donskaya-daborovikov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#interface f0/1.5
msk-donskaya-daborovikov-gw-1(config-subif)#ip nat inside
msk-donskaya-daborovikov-gw-1(config-subif)#exit
msk-donskaya-daborovikov-gw-1(config)#interface f0/1.6
msk-donskaya-daborovikov-gw-1(config-subif)#ip nat inside
msk-donskaya-daborovikov-gw-1(config-subif)#exit
msk-donskaya-daborovikov-gw-1(config)#ip access list extended natinet
^
% Invalid input detected at '^' marker.

msk-donskaya-daborovikov-gw-1(config)#ip access-list extended nat-inet
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#remark q42
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit ip host 10.129.0.200 any
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit ip host 10.129.128.200 any
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#remark sochi
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit ip host 10.130.0.200 any
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#exit
```

Figure 27: Настройка NAT на маршрутизаторе msk-donskaya-daborovikov-gw1.

Контрольная проверка.



```
pc-q42-1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.0.2.1

Pinging 192.0.2.1 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.0.2.1: bytes=32 time=1ms TTL=253
Reply from 192.0.2.1: bytes=32 time<1ms TTL=253

Ping statistics for 192.0.2.1:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.129.0.1

Pinging 10.129.0.1 with 32 bytes of data:

Reply from 10.129.0.1: bytes=32 time=6ms TTL=255
Reply from 10.129.0.1: bytes=32 time<1ms TTL=255
Reply from 10.129.0.1: bytes=32 time<1ms TTL=255
Reply from 10.129.0.1: bytes=32 time=3ms TTL=255

Ping statistics for 10.129.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 2ms
```

Figure 28: Контрольная проверка.

В ходе выполнения лабораторной работы я настроил взаимодействие через сеть провайдера посредством статической маршрутизации локальной сети организации с сетью основного здания, расположенного в 42-м квартале в Москве, и сетью филиала, расположенного в г. Сочи.